

## Introduction

### Modern Utilisation of decision trees

Decision trees remain widely used because they are fast, robust on tabular data sets. In practice, single trees are often used when stakeholders require transparent decision rules (e.g compliance heavy domains), Decision trees are widely used both as standalone models and as the base learners in ensemble methods such as Random Forests (bagging) and gradient-boosted decision trees (e.g XGBoost and LightGBM).<sup>1</sup>

- Clinical decision support / risk : rule like models can support transparent screening decisions and can be audited more easily than black-box models.
- Credit scoring and fraud detection : decision trees and boosted trees are strong on heterogeneous categorical/tabular features and capture nonlinear interactions.
- Predictive maintenance and operations : trees model interactions between sensor-derived features and failure classes with relatively low preprocessing.<sup>2</sup>

### Gini Indexes in decision trees

A decision tree selects splits that make child nodes as “pure” as possible. Gini impurity measures how mixed the classes are in a node:

$$Gini = 1 - \sum_{k=1}^K p_k^2$$

Where  $P_{vk}$  is the proportion of class K in the node. A pure node has  $Gini = 0$ . For any candidate split, the tree computes weighted impurity after splitting:

$$Gini_{split} = \frac{n_L}{n} Gini(L) + \frac{n_R}{n} Gini(R)$$

The impurity reduction (gain) is:

$$Gain = Gini(parent) - Gini_{split}$$

So training at each node the algorithm evaluates many candidates splits and picks the one that maximises impurity reduction.

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<sup>1</sup> Readthedocs.io. (2025). *Introduction to Boosted Trees — xgboost 3.2.0 documentation*. Available at: [https://xgboost.readthedocs.io/en/stable/tutorials/model.html?utm\\_source=](https://xgboost.readthedocs.io/en/stable/tutorials/model.html?utm_source=)

<sup>2</sup> scikit-learn (2025). 1.10. *Decision Trees — scikit-learn 0.22 documentation*. Scikit-learn.org. Available at: <https://scikit-learn.org/stable/modules/tree.html>.

### Dataset acquisition and preparation

The Car Evaluation dataset contains 1728 instances described by six categorical attributes (**buying**, **maint**, **doors**, **persons**, **lug\_boot**, **safety**) and a four-class target label (**unacc**, **acc**, **good**, **vgood**). A sanity check confirmed that all columns were categorical and there were no missing values. The dataset is class-imbalanced, with **unacc** comprising %70.04 (1216/1728) of examples, **acc** %22.21 (383/1728), and the minority classes **good** %3.98 (68/1728) and **vgood** 3.76% (65/1728); this is important when interpreting performance, since minority classes typically provide less training signal. For the **scikit-learn** baseline model, categorical features were converted to integer codes via label encoding and the encoding mappings were retained for transparency. Finally the data was split into **80% training / 20% test** using **stratification** to preserve class proportions and a fixed random seed for reproducibility.

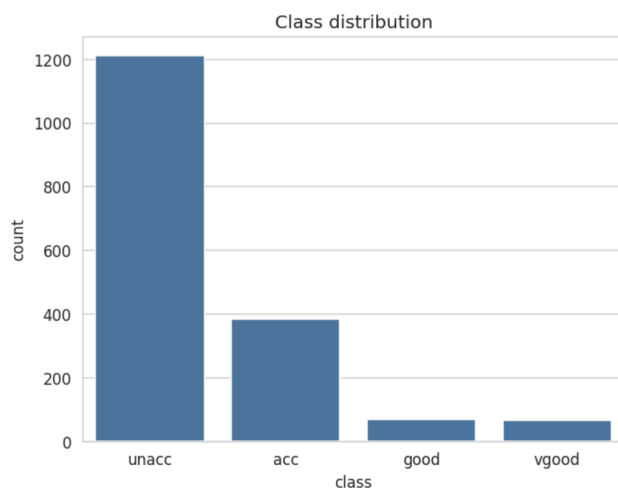


Fig1.

| class |       |       |
|-------|-------|-------|
| unacc | 70.04 | 69.94 |
| acc   | 22.21 | 22.25 |
| good  | 3.98  | 4.05  |
| vgood | 3.76  | 3.76  |

Fig2.

### Best Depth-2

A **depth-2** decision tree achieved 0.767 training accuracy and 0.778 test accuracy, indicating **limited model** capacity. In contrast, an unrestricted tree achieved 1.000 training accuracy and 0.985 test accuracy, showing that the dataset is highly separable using decision rules. The depth sweep plot shows training accuracy increases constantly with depth, while test accuracy improves rapidly and then plateaus at its maximum. The best generalization performance was achieved around **depth 12**, after which additional depth shows low improvements.

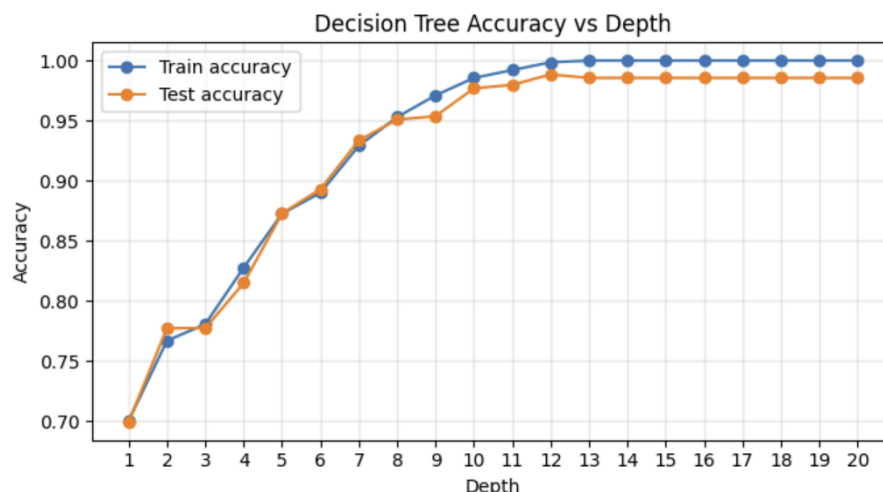


Fig3.

The confusion matrix highlights the limitations of a shallow tree. With **max\_depth=2**, the model achieves 0.78 accuracy largely by predicting the majority classes (**unacc** and **acc**), but it fails to identify minority classes: **good** and **vgood** have 0.00 recall, indicating the tree never predicts these labels. Many **acc** cases are also misclassified as **unacc** (37/77), showing the coarse decision boundaries at **depth 2**. In contrast, the unrestricted tree achieves 0.985 accuracy and correctly identifies almost all classes, including perfect recall for **good** (14/14). The remaining errors mainly involve the rare **vgood** class, which is occasionally predicted as **acc** (2/13), consistent with the smaller support for minority labels.

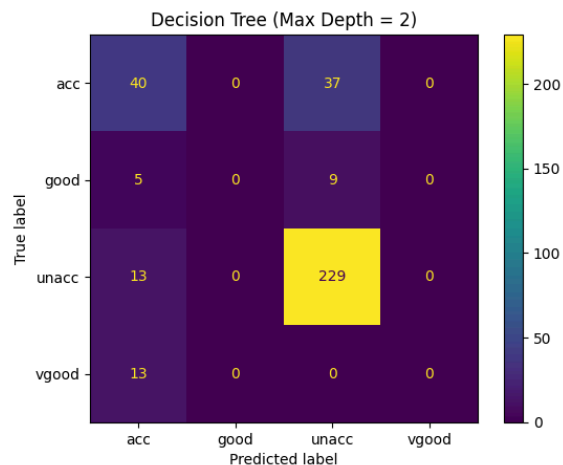


Fig4.

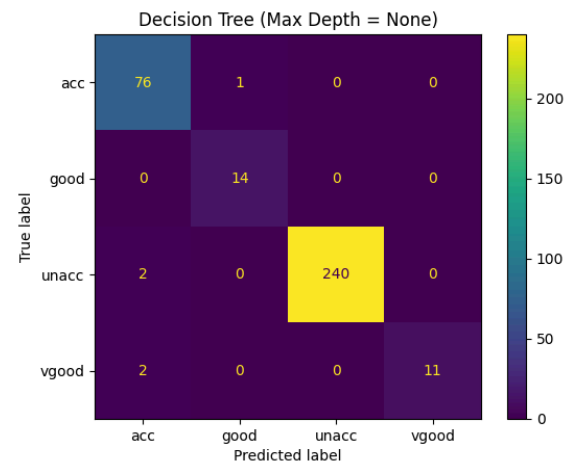


Fig5.

The **depth-2** tree produces a very small rule set driven primarily by persons and safety. When **persons = 2**, the model predicts **unacc**. For larger passenger capacity, the model predicts **acc** only when **safety = high**, otherwise it predicts **unacc**. Because this “shallow prediction tree” only contains two decision nodes, it effectively outputs only **acc** and **unacc**, explaining why **good** and **vgood** achieve zero recall in the confusion matrix (the model never predicts those classes). This demonstrates underfitting: the limited depth prevents the tree from learning the more specific combinations of attributes to separate the minority classes.

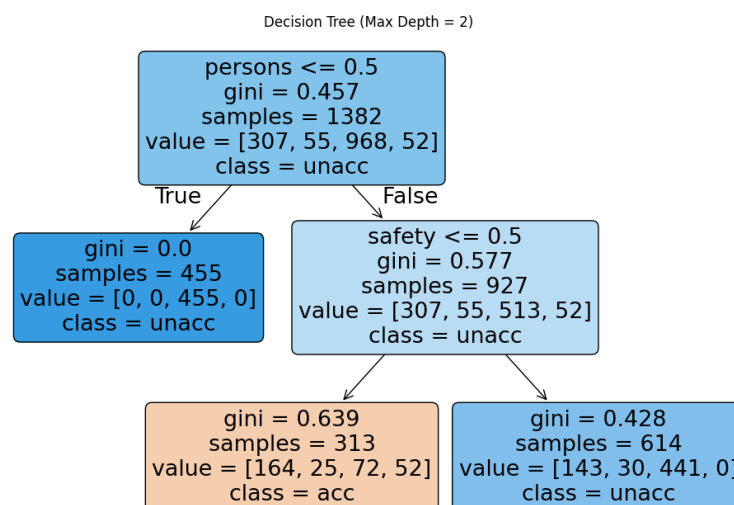


Fig6.

### Depth-2 Decision Tree

A depth-2 decision tree was implemented from scratch using NumPy/Pandas by selecting splits that minimise weighted Gini impurity. At the root node, the training data had Gini impurity 0.4570 and the best split was **persons**  $\leq 0.5$ , reducing weighted impurity to 0.3873 (gain 0.0697). The left child of this split was already pure (Gini. 0.0000), indicating all samples in that region belong to a single class, so further splitting provides no improvement (gain 0.0000). On the right child (Gini 0.5774), the best second-level split was **safety**  $\leq 0.5$ , reducing weighted impurity to 0.4988 (gain 0.0786). The resulting IF-THEN rules predict primarily **unacc**, with **acc** predicted for the region with higher passenger capacity and high safety. Evaluated on the held-out test set, from the scratch **depth-2** model achieved 0.775 accuracy, matching the **scikit-learn depth-2** baseline.

|   | node  | samples | gini_before | feature | threshold | weighted_gini_after | gain     |
|---|-------|---------|-------------|---------|-----------|---------------------|----------|
| 0 | root  | 1382    | 0.457045    | persons | 0.5       | 0.387305            | 0.069740 |
| 1 | left  | 455     | 0.000000    | buying  | 0.5       | 0.000000            | 0.000000 |
| 2 | right | 927     | 0.577406    | safet   | 0.5       | 0.498768            | 0.078639 |

Accuracy: 0.7775

Fig7.

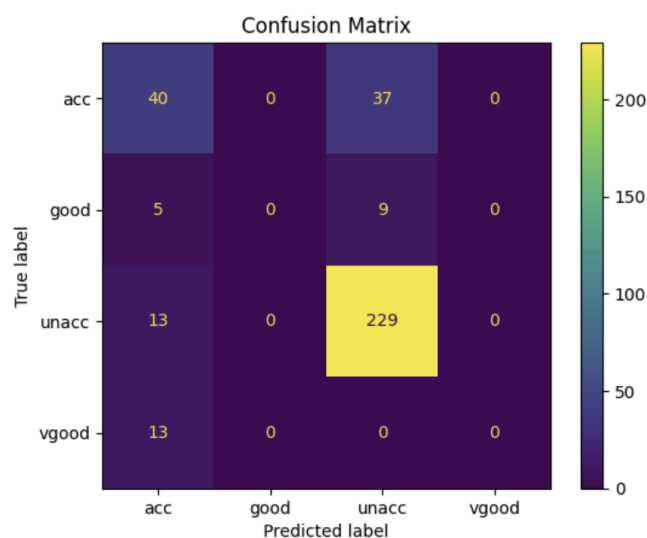


Fig8.

The from-scratch model selected the same dominant structure as the scikit-learn tree at **depth-2**: a root split on **persons** followed by a split on **safety**. This agreement supports the correctness of the manual Gini-based split search.

## Conclusion

### *Overfitting vs underfitting:*

The **depth-2** model underfits the dataset, achieving 0.767 training accuracy and 0.775 test accuracy, indicating limited capacity to represent the decision boundaries required for multi-class separation. By contrast, the unrestricted tree achieves 1.00 training accuracy and 0.9855 test accuracy, showing that the Car Evaluation dataset is highly separable using decision rules. The depth sweep demonstrates the expected bias-variance trade-off: training accuracy increases constantly with depth, while test accuracy improves rapidly and then plateaus. The best generalization performance was seen around **depth = 12**, after additional low improvements in depth.

### *Confusion matrix insights:*

Confusion matrix analysis shows that the shallow tree primarily separates the majority classes (**unacc** and **acc**) but fails to recognise minority classes. With **max\_depth=2**, both **good** and **vgood** have 0.00 recall because the tree never predicts those labels, and a substantial portion of **acc** cases are misclassified as **unacc**. In contrast, the unrestricted tree achieves strong per-class performance, correctly classifying nearly all instances, including minority classes (**good** 14/14 correct), with the remaining errors mainly involving occasional **vgood** -> **acc** confusion (2 of 13 **vgood** test samples).<sup>3</sup>

### *Comparison sklearn vs from-scratch:*

The from-scratch **depth-2** implementation selected the same dominant split structure as **scikit-learn**: a root split on **persons**  $\leq 0.5$  followed by a split on **safety**  $\leq 0.5$  in the **persons**  $> 0.5$  branch. The split log confirms impurity reduction at each decision point (root gain 0.0697, right-child gain 0.0786), and the from-scratch model achieved 0.775 test accuracy, matching the scikit-learn **depth-2** baseline. This agreement supports the correctness of the manual Gini-based search procedure.

### *Gain/Lost of from-scratch implementation:*

Implementing the tree manually improves understanding of how decision trees “search” for splits by explicitly computing Gini impurity, weighted after splitting, and impurity reduction (gain). It also gives full transparency over design choices (candidate thresholds, stopping rules, and tie-breaking). However, a from-scratch implementation is slower, easier to get wrong in edge cases. (e.g., empty splits, ties, class imbalance) and lacks optimisations/features available in mature libraries such as **scikit-learn** (intensive testing).

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<sup>3</sup> scikit-learn (n.d.). *sklearn.metrics.ConfusionMatrixDisplay*. scikit-learn. Available at: <https://scikit-learn.org/stable/modules/generated/sklearn.metrics.ConfusionMatrixDisplay.html>.

### Thoughtful and relevant justifications:

In Healthcare settings, performance is often the pinnacle of importance. A shallow tree provides a compact set of IF-THEN rules that clinicians can validate against domain knowledge, communicate to patients, and audit for safety and fairness. This supports governance requirements and reduces the risk of relying on highly specific interactions that may not generalise under dataset shifts. Therefore, even if deeper trees achieve higher accuracy, a shallow tree may be preferred when transparency and decision justification are critical.<sup>4</sup>

### Pure node stopping rule and gain analysis:

A stopping rule was added to the from-scratch decision tree to prevent splitting when a node is already pure (Gini = 0). In the original implementation, one child node after the root split was perfectly pure, resulting in a redundant split with **gain = 0**. After applying the stopping rule, this unnecessary split was removed, leaving only two meaningful splits: the root split on **persons**  $\leq 0.5$  (gain 0.069740) and the right child split on **safety**  $\leq 0.5$  (gain 0.078639). The test-set accuracy remained 0.775, showing that the removed split did not contribute to the predicted performance and improving clarity of the learned rule set.

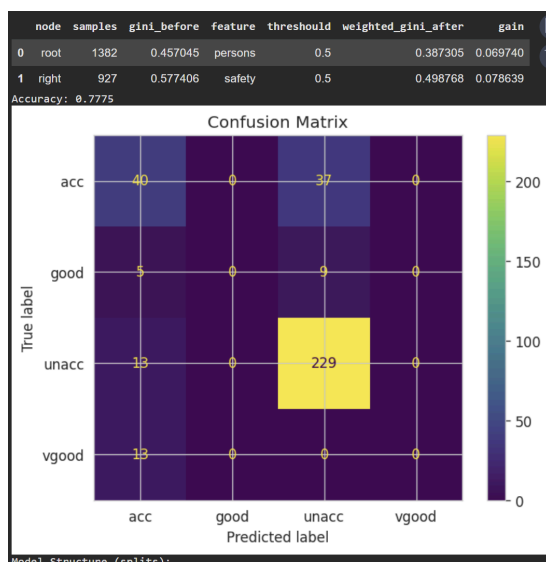


Fig9.

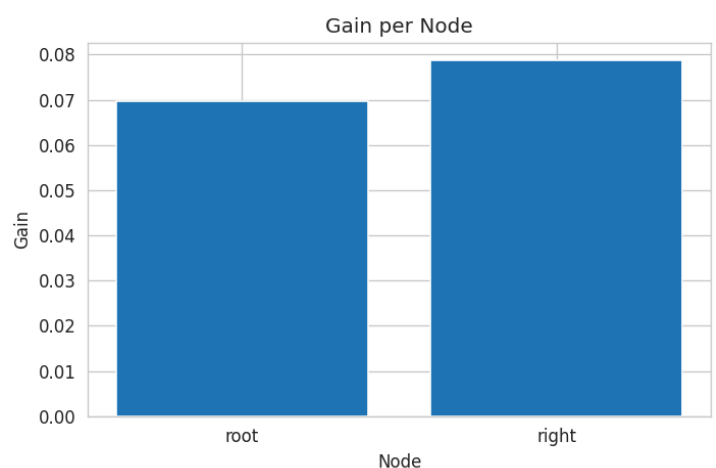


Fig10.

<sup>4</sup> Microsoft Research. (2016). *Intelligible Machine Learning Models for HealthCare - Microsoft Research*. Available at: <https://www.microsoft.com/en-us/research/video/intelligible-machine-learning-models-for-healthcare/>